25900650 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA202 - DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. If the joint pmf is p(0,0)=0.2, p(0,1)=0.3, p(1,0)=0.2, p(1,1)=0.3, find P(X=0|Y=0). [A] / [CO1]
2. If P(x, y) = cx2y; for x = 1, 2 y = 1, 2, 3 is a joint probability mass function, find c. [U] / [CO1]
3. The IQ scores in a population are normally distributed with a mean of 100 and a standard deviation of 15. Calculate the percentage of people whose IQ lies between 112 and 135. [An] / [CO2]
4. Give two practical uses of Student's t-distribution in statistics. [A] / [CO2]
5. Can the F-distribution take negative values? Why or why not? [U] / [CO2]
6. State one advantage and one disadvantage of the method of moments. [U] / [CO3]
7. Distinguish between estimator and estimate. [U] / [CO3]

8. What is meant by a composite hypothesis? [U] / [CO4]

9. What assumptions are made in the Chi-Square test for homogeneity? [U] / [CO4]

10. Give the test statistic in the case of small sample test to test the population mean when the population standard deviations are unknown. [U] / [CO4]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. The joint probability mass function is given by $f(x,y) = \frac{x+y}{21}, x = 1,2,3; y = 1,2.$ Are the variables X and Y independent? [U] / [CO1]

12. Let $U \sim \chi^2(1)$ and $V \sim \chi^2(1)$ be independent. Find the distribution, pdf, MGF, mean and variance of U+V. [A] / [CO2]

13. Define sampling distribution. If x1, x2, ..., xn are observations from a sample of population, then generate a statistic and suggest its sampling distribution. [K] / [CO2]

14. What does it mean for an estimator to be consistent? How is this property useful? State the sufficient conditions for consistentcy. [U] / [CO3]

15. In 90 spins of a roulette wheel with 18 red, 18 black, and 2 green slots, red occurs 52 times. Test whether the wheel is fair in terms of red outcomes? [A] / [CO4]

16. Briefly discuss ANOVA. If the Between Sum of Squares (SSC) is 123.33 and the Within Sum of Squares (SSE) is 94 for data collected on the test scores of students taught using three different teaching methods, conduct a one-way ANOVA to test whether there is a significant difference in the average scores among the teaching methods, assuming that the data was collected from 5 students per group. [E] / [CO4]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. Define the joint probability mass function and explain its two main properties. A bag contains one red, one green, and one blue ball. Two balls are drawn without replacement. Let X be the number of red balls drawn and Y the number of green balls drawn. Construct the joint probability mass function and distribution function. [U] / [CO1]

18. Explain the large sample tests for population mean. A sample of 100 students is taken from a population with heights having s.d. 10 cm. the mean height of the sample of students is found to be 168.8cm. can we accept the assumption that the mean height of the students is 170 cm? [A] / [CO4]

[12x1 = 12]